

Sam Nguyen

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EDUCATION

Iowa State University

Bachelor of Science, Mechanical Engineering, 4.00 GPA - President's List 2x

Expected Graduation: May 2029

Ames, Iowa

WORK EXPERIENCE

Thingz

May 2026 – Aug. 2026

Mechanical Engineering Intern

Galena, Illinois

- Designed and managed large electrical substations in Autodesk Inventor by aligning and interpreting 30M+ point LiDAR scans to provide a digital twin representative of real-world equipment and environment
- Developed accurate and detailed models of electrical substation equipment, producing 1,000+ parts and sub-assemblies
- Rapidly created and organized a library of standardized substation parts and assemblies to streamline the CAD modeling process, reducing design turnaround time by roughly 70% and providing reference models for future projects
- Operated 360° LiDAR scanners and cameras to capture point cloud data across entire 5+ acre substation on-site
- Worked cross-functionally to coordinate modeling efforts and deliver weekly verbal and written progress reports
- Consistently adhere to modeling principles and requirement standards to ensure compatibility between new and existing parts, assemblies, and naming conventions

EXTRACURRICULAR

Formula SAE Electric Vehicle (FSAE EV)

Sep. 2025 – Present

Suspension Subsystem Member

Iowa State University

- Independently designed and manufactured a static strain gauge calibrator, involving statics calculations, hands-on tooling experience, and welding, culminating in significant data acquisition for the team's R&D of strain gauges
- Created and implemented various 3D-printed jig parts to ensure the precise location of critical suspension component geometries during manufacturing such as the A-Arms, push rods, and tabs, on all 4 corners of the vehicle
- Oversee and own the bell crank/rocker assembly of the latest design cycle of the vehicle, utilizing design tables and proper modeling practices to execute and validate decisions while also improving knowledge transfer
- Researched and designed various solutions to connect suspension blade components to the anti-roll bar
- Collaborated with external design teams to ensure seamless integration between suspension parts in ARB assembly
- Analyzed bearing specs to determine the best solution for use case, considering packaging conflicts and costs
- Precisely 3D modeled parts to be easily scalable through use of parametric modeling techniques

SolidWorks Club

Aug. 2025 – Present

Sophomore Representative

Iowa State University

- Achieved SolidWorks CSWP Certification through weekly CAD assignments and rigorous final 3-hour assessment which tested part modeling, design iteration, and assembly skills, scoring a flawless 100%
- Evaluated engineering drawings of models and assemblies in accordance with ASME Y14.5 standards
- Elected as Sophomore Representative after first year of membership demonstrating rapid growth and mentorship
- Assisted club members in learning core SolidWorks skills by providing 1-on-1 feedback and first-hand experience
- Demonstrated excellent interpersonal communication skills to achieve a record of 163 new member sign-ups at info booth

PROJECTS

Fishing Pole Holder

Sep. 2025 – Nov. 2025

- Assessed design requirements for the product to ensure proper integration, using digital calipers and paper sketches
- Developed and manufactured product design using SolidWorks and Bambu Lab X1C, material choice of PETG

SKILLS AND CERTIFICATIONS

SolidWorks CSWP, Autodesk Inventor, 3D Printing (FDM), Machining, Python, Microsoft Office (Excel, Word, PowerPoint)

INTERESTS: Fishing, Sketching, Drafting Thoughts, Filmmaking, Digital Cameras, Soccer, Automotive, Astronomy